Properties of Discrete-time Fourier series (DTFS)

$$x[n] \xleftrightarrow{FS} a_k$$
$$y[n] \xleftrightarrow{FS} b_k$$

① Linearity: $Ax[n] + By[n] \xleftrightarrow{FS} Aa_k + Bb_k$

② Time-shift: $x[n-n_0] \xleftrightarrow{FS} a_k e^{-jk(2\pi/N)n_0}$

<u>Proof:</u> $b_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n-n_0] e^{-jk(2\pi/N)n}$

Let $n - n_0 = p \Rightarrow n = p + n_0$

$$\therefore \quad b_k = \frac{1}{N} \sum_{p=\langle N \rangle} x[p] e^{-jk(2\pi/N)(p+n_0)}$$

$$= e^{-jk(2\pi/N)n_0} \frac{1}{N} \sum_{p=\langle N \rangle} x[p] e^{-jk(2\pi/N)p}$$

$$= e^{-jk(2\pi/N)n_0} a_k$$

In a similar way other properties can also be proved (analogous to the CT case).

③ Conjugation: $x^*[n] \xleftrightarrow{FS} a^*_{-k}$

④ Frequency-shift: $e^{jM(2\pi/N)n} x[n] \xleftrightarrow{FS} a_{k-M}$

⑤ Multiplication: $x[n]y[n] \xleftrightarrow{FS} \underbrace{\sum_{l=\langle N \rangle} a_l b_{k-l}}_{\text{periodic-convolution}}$

⑥ <u>Periodic Convolution:</u> (Circular convolution)

This is something new that we have never covered. So, convolution between two discrete signal is defined as:

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]\, y[n-k]$$

But periodic convolution has the sum different.

$$x[n] \circledast y[n] = \sum_{k=\langle N \rangle} x[k]\, y[(n-k) \bmod N]$$

$$x[n] \circledast y[n] \xleftrightarrow{FS} N a_k b_k$$

(The proof of this is your assignment)

⑦ Parseval's Relation for periodic signal

$$\frac{1}{N}\sum_{n=\langle N\rangle}|x[n]|^2 = \sum_{k=\langle N\rangle}|a_k|^2$$

As in the continuous-time case, the left hand side of Parseval's relation is the average power in one period of the periodic signal $x[n]$. Similarly, $|a_k|^2$ is the average power in the k-th harmonic component of $x[n]$.

∴ Parseval's relation states that the average power in a periodic signal equals the sum of the average powers in all its harmonic components.